Group	Descriptive feature name	Datafile column name	Image type	Input	Value normalization	Calculated per	Threshold	Description
Basic biofilm metrics	Biomass	Biomass_um3_per_um2	3D	Mask	No	Whole image	Applied threshold	Each voxel marked as biofilm in the mask (after thresholding or from an imported mask) is multiplied by the voxel volume (voxel dimensions are provided to the software) to give the biomass volume, which is then divided by the substratum (field) area. This indicates how much biofilm is present per unit surface area.
Basic biofilm metrics	Biomass volume	BiomassVolume_um3	3D	Mask	No	Whole image	Applied threshold	The number of voxels marked as biofilm in the mask multiplied by the voxel volume (dx, dy, dz – the voxel size you set in the software), reported in cubic micrometres. It measures the total amount of biofilm material in the imaged volume, and a higher value means more biofilm material.
Basic biofilm metrics	Maximum biofilm height	MaxBiofilmHeight_COMSTAT_um	3D	Mask	No	Whole image	Applied threshold	The software identifies the highest z-layer containing any biomass and multiplies the layer's index plus one (as layers are counted from 0 to obtain the correct layer number) by the z-step. It thus measures how tall the biofilm is at its single highest point. A higher value means the biofilm extends further up from the substratum. As it is based on a single point, it is also sensitive to noise.
Basic biofilm metrics	Biomass height (95th percentile)	BiomassHeightP95_um	3D	Mask	No	Whole image	Applied threshold	The software counts biomass voxels in each z-layer of a 3D image, sums them from the substratum upwards, identifies the layer at which 95% of the total biomass has been reached, and multiplies (that layer + 1) by the z-step to obtain a measure in µm. It measures how high the biofilm extends – the height below which 95% of the biomass lies. A higher value means the bulk of the biofilm extends further upwards, and vice versa.
Basic biofilm metrics	Mean thickness (biomass columns)	MeanThickness_BiomassColumns_um	3D	Mask	No	Over occupied columns	Applied threshold	For each x–y position (column), the software counts the z-steps that contain biomass (not necessarily connected) and multiplies this by the z-step size (set in the software) to give that column's filled thickness. The mean is calculated over occupied columns only (empty columns are excluded), giving the average thickness of the biofilm where it is present. It measures filled biomass – the voxels that actually contain biofilm – rather than the distance from the substratum to the top of the column. Because the average is calculated over occupied columns only, this value is always ≥ biomass (the same filled thickness averaged over the whole field).
Basic biofilm metrics	Max compacted column thickness	MaxCompactedColumnThickness_um	3D	Mask	No	Whole image	Applied threshold	For each x–y position (column), the software counts the biomass voxels in that column and multiplies by the z-step to obtain the column's filled ("compacted") thickness; the feature then reports the maximum across all columns. Internal gaps are removed, and the biomass voxels do not need to be connected. It represents the filled height of the column containing the most biomass. A higher value means that at least one column is very densely filled with biomass.
Basic biofilm metrics	Roughness coefficient	Roughness_RaStar	3D	Mask	No	Whole image (incl. empty columns)	Applied threshold	For every x–y column of the image, the software takes the filled thickness (biomass voxels × z-step, with empty columns included as zero), computes the mean absolute deviation of these thicknesses (|thickness − mean thickness|), and divides this by the mean thickness (count × z-step). The result is dimensionless and measures how uneven the biofilm is. A value of 0 means an even biofilm (every column has the same thickness); a higher value indicates more variation between columns. Because empty areas count as zero-thickness valleys, a patchy or sparsely covered biofilm also scores as rough.
Basic biofilm metrics	Substratum coverage	SubstratumCoverage_pct	2D, 3D	Mask	No	Substratum layer	Applied threshold	The software takes the bottom z-layer (the first layer of the 3D image, i.e. the substratum), counts the pixels in it that are biomass, divides this by the total number of pixels in that slice (image width × height), and then multiplies by 100. It measures how much of the attachment surface is covered by biofilm; the first slice must correspond to the substratum plane and be in focus – if it is below or above the true attachment surface, coverage is under- or overestimated. A higher value means more of the surface is colonised at the base.

Group	Descriptive feature name	Datafile column name	Image type	Input	Value normalization	Calculated per	Threshold	Description
Basic biofilm metrics	Layer area	LayerArea_um2	3D	Mask	No	Per layer (summarized)	Applied threshold	For each z-layer of the 3D image, the software counts the biomass pixels in the mask. Layer area multiplies that count by the pixel area (dx × dy) to give the biomass area at that depth in µm². Layer occupancy instead divides the count by the total number of pixels in the layer and multiplies by 100, giving the percentage of that depth slice filled with biomass. Both are per-layer profiles across the stack; the SUMMARY then reduces each profile to descriptive statistics: mean/median – the typical biomass area/occupancy across depth; min – the sparsest layer (often an empty top layer = 0); max – the densest layer (usually near the substratum); q10/q25/q75/q90 — the spread of the biomass profile; std/var – how much biomass varies between layers: high = concentrated in some depths (base-heavy), low = more uniform through depth. Thus std/var of occupancy is essentially a measure of vertical heterogeneity. These values are computed over all layers, including empty top slices (= 0), so the mean is pulled down by empty layers.
Basic biofilm metrics	Layer occupancy	LayerOccupancy_pct	3D	Mask	No	Per layer (summarized)	Applied threshold	
Basic biofilm metrics	Biomass centre-of-mass height	BiomassCenterOfMassHeight_um	3D	Mask	No	Whole image	Applied threshold	For each z-layer, the software multiplies the layer height by the biomass in that layer, sums over all layers (layer index × z-step × biomass of that layer), and divides by the total biomass. It represents the vertical centre of the biofilm by calculating the average height at which the biomass sits. A low value means the biomass is concentrated near the substratum; a high value means it sits higher up. It describes where the mass is centred vertically, not how high the biofilm extends.
Basic biofilm metrics	Biomass vertical spread	BiomassVerticalSpread_um	3D	Mask	No	Whole image	Applied threshold	Using the same per-layer weighting as the centre of mass, the software takes each layer's squared distance from the centre-of-mass height, weights it by the biomass in that layer, averages over the total biomass, and takes the square root – the biomass-weighted standard deviation of voxel heights, sqrt(Σ (layer_height − z_com)² × layer_biomass / total_biomass). It measures how vertically spread out the biomass is around its centre of mass (how wide the vertical distribution is). A small value means the biomass occupies a narrow band of heights (a thin layer); a large value means it is distributed across many layers.
Basic biofilm metrics	Spatial spreading (horizontal / vertical / total)	SpreadingHorizontal SpreadingVertical SpreadingTotal	3D	Mask	No	Whole image	Applied threshold	The software takes every biomass voxel's position, converts it to µm, and measures how dispersed those positions are along each axis by calculating the standard deviation of the voxel coordinates about the biomass centroid (the per-axis mean position). This centroid is the centre of biomass, not the image centre: if microcolonies lie towards the edge, the centroid shifts there too. Horizontal = spread across the x–y plane (lateral extent). Field-size dependent, and does not distinguish full coverage from scattered patches. Vertical = spread in height = standard deviation of the voxel z-positions about their mean. Identical to BiomassVerticalSpread. Total = combined 3D spread = √(horizontal² + vertical²) = √(var_x + var_y + var_z); the overall 3D dispersion of the biomass. Large = biomass dispersed across the image (broad coverage or patches far apart); small = biomass concentrated in one region.

The substratum is the surface to which the biofilm attaches. In 3D images, this is usually the first layer, and in 2D images it is usually the only layer. Questions regarding substratum colonisation typically concern how much surface is covered; whether the biofilm consists of many separate microcolonies or a confluent lawn; the size and spacing of these biomass components; and whether the attached layer already contains empty space and what this looks like. These features are therefore calculated for both 2D and 3D images.

For a 3D stack, the software assumes the first layer of a 3D image is the substratum and computes all substratum features on that layer only. Objects are taken to be 8-connected (pixels touching by edge or corner); this is fixed for the substratum slice of a 3D stack and is the default for 2D images. The features mirror the 3D object/pore features described above but are computed in the plane of a single layer, so area replaces volume and circles replace spheres. The same computations are applied to genuine 2D images, in which the single layer is usually taken to represent the substratum.

Group	Descriptive feature name	Datafile column name	Image type	Input	Value normalization	Calculated per	Threshold	Description
Basic biofilm metrics	Substratum biomass area	SubstratumBiomassArea_um2	2D, 3D	Mask (bottom layer)	No	Substratum layer	Applied threshold	The software calculates the biomass-occupied pixels multipled by pixel area provided in the software. It measures the total biomass area on the substratum layer. It is the companion feature to SubstratumCoverage_pct, which expresses the same measure as a percentage of the layer.
Basic biofilm metrics	Substratum object count	SubstratumObjectCount	2D, 3D	Mask (bottom, 8-connectivity)	No	Substratum layer	Applied threshold + min size	This is the 2D analogue of ObjectCount3D. It measures the number of separate 8-connected (default setting) biomass regions on the substratum. For 2D images these are the objects in objects_rgb; for 3D stacks they are re-labelled independently on the bottom layer and may not match the 3D objects_rgb. It is most informative when objects are well separated; a cluster of touching cells counts as one object. Read with SubstratumObjectArea for reliable interpretation. It is threshold- and connectivity-dependent, and minimum-size dependent.
Basic biofilm metrics	Substratum object density	SubstratumObjectDensity_per_um2	2D, 3D	Mask (bottom)	No	Substratum layer	Applied threshold + min size	This is a 2D analogue of ObjectDensity3D, calculated by dividing object count by substratum area. It is comparable across images with the same field size.
Basic biofilm metrics	Substratum object area (distribution)	SubstratumObjectArea_aggregations_um2	2D, 3D	Mask (bottom)	No	Per object (summarised)	Applied threshold + min size	The 2D analogue of ObjectVolume3D, where biomass pixel counts are multiplied by the pixel area and summary of the object properties is given by aggregation of calculated values.
Basic biofilm metrics	Substratum object equivalent diameter (distribution)	SubstratumObjectEquivalentDiameter_aggregations_um	2D, 3D	Mask (bottom)	No	Per object (summarised)	Applied threshold + min size	Each object's area is re-expressed as the diameter of a circle of equal area ($d = 2 \cdot \sqrt{(area/\pi)}$), summarised as a distribution. This feature helps to intuitively understand the distribution of object sizes on the substratum. Because aggregates and microcolonies are irregular, each object's area is converted into the diameter of a circle of equal area (not a circle drawn around it), providing an intuitive length.
Basic biofilm metrics	Substratum object nearest-neighbour (distribution)	SubstratumObjectNearestNeighbor_aggregations_um	2D, 3D	Mask (bottom)	No	Per object (summarised)	Applied threshold + min size	The 2D analogue of ObjectNearestNeighbor3D, describing the distribution of centroid-to-centroid distances from each object to its nearest neighbour on the substratum. Small values mean biomass parts are clustered; large values mean biomass parts are spread apart. Centre-to-centre (ignores object size; not the edge-to-edge gap); requires ≥2 objects (otherwise NaN).
Basic biofilm metrics	Substratum internal pore count (2D)	SubstratumInternalPoreCount2D	2D, 3D	Mask (bottom)	No	Substratum layer	Applied threshold	The 2D analogue of InternalPoreCount3D, referring to the number of enclosed holes (pores) in the substratum biomass – empty space that is completely surrounded by biomass within the layer. Gaps that reach the layer border (open to the outside) are excluded, so it counts only sealed holes, not open gaps at the edge. Sensitive to segmentation.
Basic biofilm metrics	Substratum internal porosity (2D)	SubstratumInternalPorosity2D	2D, 3D	Mask (bottom)	No	Substratum layer	Applied threshold	The 2D analogue of InternalPorosity3D, in which enclosed pore area is divided by the biofilm body area (biomass area plus enclosed pore area), with open (border-connected) gaps excluded. The higher the value, the more internally perforated the biofilm.
Basic biofilm metrics	Substratum internal pore area (distribution)	SubstratumInternalPoreArea_aggregations_um2	2D, 3D	Mask (bottom)	No	Per pore (summarised)	Applied threshold	This feature describes the population of internal pores determined for SubstratumInternalPoreCount2D. It summarises the distribution of individual enclosed-pore area, calculated by multiplying the pixel count by the pixel area. It characterises pore size and its variability (typical pore from mean/median, heterogeneity from Std/IQR, large pores from P90), complementing the count (how many) and porosity (how much of the objects is empty space). It analyses enclosed pores only and is NaN when there are no pores.
Basic biofilm metrics	No biomass detected (flag)	NoBiomassDetected	3D	Mask	No	Whole image	Applied threshold	Quality flag to check the segmentation: the software returns true, meaning that the threshold has detected no biofilm at all, indicating that you need to check the segmentation.

Group	Descriptive feature name	Datafile column name	Image type	Input	Value normalization	Calculated per	Threshold	Description
3D object features	3D object count	ObjectCount3D	3D	Mask (26-connectivity)	No	Whole image	Applied threshold + minimum object size	ObjectCount3D counts the separately coloured regions in the _object_labels_rgb output (check that file in results/feature_generator/segmentation), where each region is a group of connected biomass voxels. Each object's volume is its voxel count multiplied by the voxel volume set in the software (e.g. dx·dy·dz = 0.13·0.13·0.5 = 0.00845 µm³). Objects below a user-set minimum volume are discarded (default 0 µm³); a biologically motivated cut-off is approximately half a single-cell volume, estimated from cell dimensions measured in ImageJ (e.g. sphere $(4/3)\pi r^3$ for cocci, cylinder $\pi r^2 h$ for rods, measuring r and h with ImageJ tools). "Connected" (you set the connectivity in the software) is a voxel-neighbour rule defining how two neighbouring voxels connect: 6 (shared whole face), 18 (face/edge), or 26 (face/edge/corner; default in µICS). Higher connectivity merges more and yields fewer, larger objects. The count can be misleading when the mask does not resolve down to single cells, so judge it against the per-object volume distribution (ObjectVolume3D Mean/Median/Std/IQR): many objects with tiny volumes = dispersed single cells; few objects with large volumes = microcolonies; a spread = a mix. To see the situation directly, refer to the mask and _object_labels_rgb.
3D object features	3D object density	ObjectDensity3D_per_um3	3D	Mask	No	Whole image	Applied threshold + min size	The software takes the 3D object-count feature and divides it by the imaged volume (nx·ny·nz × voxel volume, e.g. 21·1024·1024 × 0.00845 ≈ 1.86×10^5 µm³). It reports how many objects there are per µm³ of the entire imaged volume (not per volume of biomass). It is sensitive to acquisition depth and field dimensions: because the total image volume is used and much of it is empty space above the biofilm, the objects are, on average, very diluted. It can therefore be compared only with images of the same depth and field size. It also inherits all the 3D object-count caveats.
3D object features	3D object volume (distribution)	ObjectVolume3D_aggregation_um3	3D	Mask	No	Per object (summarised)	Applied threshold + min size	The volume of each 3D object (voxel count × voxel volume) is computed for every measured object in the image, then summarised across objects. This describes the distribution of object volumes. Mean and median indicate the typical object size (a mean above the median signals a few large colonies skewing the average); standard deviation and interquartile range quantify the spread and heterogeneity of sizes; P90 highlights the larger structures. Together with ObjectCount3D, this helps to characterise the size of the detected objects and enables separation of many small objects (dispersed single cells) from a few large ones (microcolonies). Objects below the user-set minimum volume are already excluded, so these statistics summarise the same object set as the count.
3D object features	3D objects nearest-neighbour distance (distribution)	ObjectNearestNeighbor3D_aggregation_um	3D	Mask	No	Per object (summarised)	Applied threshold + min size	For each 3D object, the distance from its centroid (mean voxel position, the centre of the object) to the nearest other object's centroid is measured as a true physical distance in µm. The measurement is anisotropic, with each axis scaled by its own voxel size (x, y × 0.13, z × 0.5). These distances are summarised across objects. Small distances indicate objects clustered together; large distances indicate they are spread apart. Interpret these measures alongside ObjectCount3D and ObjectVolume3D to understand the full spatial pattern. Note that this is a centre-to-centre measure and ignores object size (it is not the edge-to-edge gap), requires at least two objects (otherwise NaN), and inherits the object definition set by threshold and connectivity.

Group	Descriptive feature name	Datafile column name	Image type	Input	Value normalization	Calculated per	Threshold	Description
Advanced 3D geometry)	Surface area	SurfaceArea_voxel_faces_um2	3D	Mask	No	Whole image	Applied threshold	A wrap around the objects (around any shape of object, e.g. a cylinder or sphere; and if it is hollow, around the hollow too; if it is porous, around the pores, so it reads as a cylinder-with-a-pore). It is computed on the whole biomass mask as a single total, with every exposed voxel face added into one µm² value, not as a per-object distribution. Scroll _object_labels_rgb / the mask slices to picture the shape driving it, ignoring the colour. It is a staircase (voxel-face) estimate that overestimates smooth surfaces, and it combines the outer surface with internal pore walls. For interpretation, pair it with the surface-to-volume ratio and the pore features.
Advanced 3D geometry	Surface-to-volume ratio	SurfaceToVolumeRatio_um_inv	3D	Mask	No	Whole image	Applied threshold	This is the ratio of two pre-calculated features, SurfaceArea_voxel_faces_um2 ÷ BiomassVolume_um3, measuring the amount of exposed surface per unit biomass. If the biomass is irregular – full of holes, channels, or thin spanning shapes – the surface is large relative to its volume and the ratio is high, implying an irregular, porous, channelled structure (or finely divided, thin biomass, since the ratio scales as 1/size). A compact, smooth blob with no channels gives a low ratio.
Advanced 3D geometry	Mean diffusion distance	MeanDiffusionDistance_um	3D	Mask	No	Whole image	Applied threshold	For every biomass voxel in the image, the 3D Euclidean distance transform gives the straight-line distance to the nearest empty-space voxel (where there is no biomass). MeanDiffusionDistance_um is the average of this distance over all biomass voxels. It measures how far biomass voxels are from the nearest empty space, indicating whether cells are buried in thick structures (such as microcolonies) or sit in thin structures (such as small aggregates). A large solid blob (thick in all three dimensions) gives large distances, meaning many bacteria are buried deep inside. A hollow or channelled blob gives smaller distances because the internal voids are themselves nearby empty space, making the structure likely more penetrable and more exposed. Many small or thin objects also give small distances. The distance is set by the nearest “edge” in any direction, so it reflects local thickness and ignores overall dimensions (for example, a long thin object and a small compact clump with the same smallest thickness give similar values).
Advanced 3D geometry	Maximum diffusion distance	MaxDiffusionDistance_um	3D	Mask	No	Whole image	Applied threshold	The diffusion distances previously calculated for MeanDiffusionDistance (summarised there by the mean) are summarised here by the maximum reffering to the single most deeply buried point in the biomass that is the voxel farthest from any empty space. Because it reflects just one point, it is sensitive to outliers or segmentation artefacts and is therefore a less robust measure than the mean.
Advanced 3D geometry	Internal pore count (3D)	InternalPoreCount3D	3D	Mask	No	Whole image	Applied threshold	The software inverts the image so biomass becomes background and empty space becomes foreground. It runs connected-component analysis on the empty space (rule 26 by default), assigning each empty region a label. It then checks which labels appear on the outer sides of the image, meaning the first and last slice along each axis (the six faces of the box). Regions touching the edges of the image are “truly empty” external space; regions that never reach them are internally trapped voids, i.e. pores, and the size/count calculations are based on those. The feature reports the pore count.
Advanced 3D geometry	Internal porosity (3D)	InternalPorosity3D	3D	Mask	No	Whole image	Applied threshold	The software sums all pore volumes and divides this by the combined biomass and pore volumes. This feature reports the enclosed void fraction of the biofilm body (the volume of biomass and pores), providing an estimate of how porous (internally hollow) the biofilm is.

Group	Descriptive feature name	Datafile column name	Image type	Input	Value normalization	Calculated per	Threshold	Description
Advanced 3D geometry	Internal pore volume (distribution)	InternalPoreVolume{Mean,Median, Std,IQR,P90}_um3	3D	Mask	No	Per pore (summarised)	Applied threshold	This feature addresses the population of internal pores determined for InternalPoreCount3D. It summarises the distribution of individual enclosed-pore volumes (each = voxel count × voxel volume). It characterises pore size and its variability (typical pore from mean/median, heterogeneity from Std/IQR, large pores from P90), complementing the count (how many) and porosity (how much total void). Enclosed voids only; NaN when there are no pores, and Std/IQR = 0 with a single pore.
Advanced 3D geometry	Local biomass density	LocalBiomassDensity_aggregation + LocalBiomassDensityRadius_um (fixed to 2 micrometers)	3D	Mask	No	Per biomass voxel (summarised)	Applied threshold	This feature is based at the voxel level like the "diffusion distance" feature. Each biomass voxel is analysed with a sliding window: an invisible ball of ~2 µm radius (configurable in the code) is centred on it and moved voxel by voxel. The fraction of that ball occupied by biomass is computed for each voxel and edge-corrected for how much of the ball lies inside the image, with convolution performing all the counts efficiently, and the per-voxel fractions are summarised. It reflects local compactness: thin or isolated biomass gives low values; locally packed, solid biomass (dense microcolonies or a dense sheet) gives high values, independent of how much of the image is covered.
Advanced 3D geometry	Local thickness	LocalThickness_aggregation_um + LocalThicknessRadiusStep_um (it records the radius-sampling step used in the local-thickness computation)	3D	Mask	No	Per biomass voxel (summarised)	Applied threshold	The software performs the same distance-to-nearest-empty calculation used for MeanDiffusionDistance_um and uses it to identify the possible centre voxels where a hypothetical ball of a given size fits within the biomass (covering biomass voxels only). The ball size is decreased in each round until all biomass in the image is covered. For each voxel, the largest ball that covers it is recorded and retained. The same voxel can be covered many times by smaller balls placed to cover neighbouring voxels, but this does not change its retained value. If most of the biomass is covered by large balls, the biofilm is thick; if only small balls fit, it is thin. Once every voxel holds the diameter of its largest covering ball, those diameters are summarised.
Intensity (threshold-free)	Layer intensity statistics	diff, max, med, std, mean, min, q10/q25/q75/q90, minProp	2D, 3D	Raw image (+raw/max-normalised)	Yes (normalised and per-slice Normalized variants)	Per layer	None (threshold-free)	This set of features is calculated first by layer and then summarised per image. It is based on the whole image without thresholding. The aggregations: mean – average brightness of the slice; med – median brightness (robust to a few bright outliers); max – brightest pixel; min – darkest pixel (usually background); diff – range calculated as max − min pixel intensity, referring to the slice's dynamic range/contrast; std – spread of brightness giving information on contrast/heterogeneity; q10/q25/q75/q90– the 10th/25th/75th/90th percentiles, which describe the shape of the brightness distribution; minProp – proportion of pixels below the layer mean, a skewness indicator useful for bright biofilms with dim areas or where there is a lot of empty space. The set comes in three forms: the base statistics (on 0–1 stack-normalised intensity), RawIntensity* (the same statistics on the original intensity units, preserving absolute brightness), and *Normalized (each slice's value as a percentage of the stack total, a vertical profile of how the quantity is distributed across layers). For 2D images all aggregations have the same value.
Intensity (threshold-free)	Global mean & depth differences	globalMean, mdiffs, maxdiffs, mindiffs, eigen (DiffGlobal file)	2D, 3D	Raw image (normalised)	Yes	Whole image	None (threshold-free)	globalMean is the average normalised brightness of the whole stack. mdiffs, maxdiffs and mindiffs describe how the per-layer mean intensity changes from one layer to the next through the stack – the average change, the largest increase and the largest drop respectively; a strong systematic drop can serve as an acquisition quality check. maxdiffs and mindiffs reflect single layer-to-layer transitions, so they are sensitive to one abrupt change. eigen is currently not computed (hard-coded to 1.0). For 2D images all aggregations have the same value.

Group	Descriptive feature name	Datafile column name	Image type	Input	Value normalization	Calculated per	Threshold	Description
Geometric (threshold-free)	Fractal dimension (binary)	FractalDimension	2D, 3D	Raw image, per layer (median-binarised)	No	Per layer	None (threshold-free)	The software binarises each layer at its median, then performs boundary box-counting over a range of box sizes, meaning that it counts the boxes that straddle the edge (partly filled with biomass, partly empty; completely filled boxes are left out), and plots box size against boundary box-count on a log–log scale. The slope gives a dimensionless number reflecting the complexity or roughness of the outline (~1 smooth, ~2 convoluted).
Geometric (threshold-free)	Grayscale box fractal dimension	FractalGrayscaleBoxDimension	2D, 3D	Raw image	No	Whole image	None (threshold-free)	To calculate this fractal dimension, the original greyscale intensities are used. Across a range of box sizes, the image is partitioned into blocks, and for each block the number of boxes required to span that block's local intensity range (max − min) is counted and then summed over all blocks. Plotting the countagainst box size on a log–log scale yields, as the slope of the best-fit line, a dimensionless number. It describes texture rather than geometric complexity, but still belongs in the fractal family: it captures the roughness of the intensity surface – values towards 3 indicate a rough, bumpy intensity landscape with lots of fine variation; values towards 2 indicate a smooth, gradual one. The observed ~2–3 range is consistent with this interpretation.
Geometric (threshold-free)	Homogeneity	Homogeneity	3D	Raw image (normalised)	No	Whole image	None (threshold-free)	Homogeneity is calculated from a co-occurrence matrix that captures how often each pair of intensity levels occurs between neighbouring voxels in the X, Y and Z directions (after Beyenal et al. 2004, doi: 10.1016/j.mimet.2004.08.003), computed on the normalised greyscale image without a mask. It measures how similar neighbouring voxels are in intensity, with higher values indicating a smoother, more uniform image and lower values indicating frequent sharp intensity transitions. One value is reported per image (whole stack), so the aggregated values are identical.
Geometric (threshold-free)	Haralick GLCM texture features	GLCM Contrast, Dissimilarity, Homogeneity, Energy, Correlation, Asm_d1_angle 0,45,90,135 + _mean	2D, 3D	Raw image (normalised, 32 levels)	No	Per layer (averaged)	None (threshold-free)	The raw image is reduced to a 5-bit, 32-level greyscale. For each 2D slice, a co-occurrence matrix tallies the intensity pairs of neighbouring pixels not only along the horizontal direction but at four angles (0/45/90/135°); it does not use the z-direction (unlike the Beyenal homogeneity). For each slice, the co-occurrence matrix tallies every pair of neighbouring pixels and is then normalised into a probability distribution: each cell P(i, j) is the fraction of all neighbouring pairs in which a pixel of intensity i sits next to one of intensity j (the matrix sums to 1). Every texture metric is a weighted sum of these probabilities, differing only in how the pairs are weighted: Contrast – weights pairs by the squared intensity difference $(i - j)^2$: how strongly neighbours differ; high = rough, high-contrast texture. Dissimilarity – the same idea with linear weighting $\lvert i - j \rvert$: a gentler contrast, less dominated by large jumps. Homogeneity – weights by $1/(1 + (i - j)^2)$, favouring similar pairs: how alike neighbours are; high = smooth. Energy / ASM – the sum of squared probabilities ($ASM = \Sigma P^2$; $Energy = \sqrt{ASM}$): how concentrated the distribution is. When a few pairings dominate (a uniform or repetitive, hence highly predictable texture), the probability piles into few cells and energy is high; when many pairings occur with similar probability (a varied, unpredictable texture) it is spread thin and energy is low. Energy and ASM carry identical information. Correlation – how well a neighbour's intensity can be predicted from a pixels by a linear model: high means strong linear structure or gradients.

Group	Descriptive feature name	Datafile column name	Image type	Input	Value normalization	Calculated per	Threshold	Description
Serial / multi-threshold (ML ensemble)	Counts across thresholds	counts(inten<n), counts(Norminten<n) (+ *Normalized)	3D	8-bit image (and per-layer normalised)	Yes (Norm and Normalized variants)	Per layer	0.01-0.29 (step 0.01)	These features are calculated across a range of thresholds, independently of the optimal threshold. The software gradually raises a threshold; at each cutoff from 0.01 to 0.29 (in 0.01 steps, autoscaled to layer's own peak brightness), the layer is thresholded and its 8-connected bright regions are counted. Counts are obtained for each layer and then summarised per image. It uses two different inputs: "inten" uses the intensity image directly, while "Norminten" uses a per-layer, mean-centred, range-scaled image, which cancels layer-to-layer brightness differences so the sweep reflects pattern rather than absolute brightness. The "Normalized" variant then expresses each layer's count as a percentage of that count column's total across all layers (× 100). Because touching cells merge and the counts depend heavily on the threshold, these are best read as a texture/fragmentation signature for machine learning, not as literal cell counts.
Serial / multi-threshold (ML ensemble)	Biomass_um3_per_um2 / thickness / roughness/ LayerArea_um2 across thresholds	BioVolumeThr{n}, ThicknessThreshold={n}, RoughnessThreshold={n}	3D	8-bit image	No	Whole image	0.01-0.29 (step 0.01)	Similarly to the counts, these features are calculated using a threshold sweep to provide rich information for machine learning. Rather than using the chosen optimal threshold, which is the most intuitive way to analyse images, the software varies the threshold from 0.01 to 0.29 (in 0.01 steps, as an absolute fraction of the 8-bit range), and at each cutoff re-thresholds the entire 3D stack and recomputes one of the features. Each value is therefore a single whole-image measurement at that cutoff. The features calculated across the thresholds are Biomass_um3_per_um2 (the biomass volume per unit substratum area), MeanThickness_BiomassColumns_um (the mean filled thickness over occupied columns), LayerArea_um2 and Roughness_RaStar (the mean absolute deviation of per-column filled thickness from the field mean, normalised by that mean).